

# JAWAN PAKISTAN ASSIGNMENT #1

## Python

AMJAD ALI KHAN  
Batch -II  
Backend Development

```
class-2.py X
C: > Users > amjad > Documents > Python-JP > class-2.py > ...
1  # Assignment #1 Amjad Ali Khan 23/05/2024
2
3  # problem 1
4  """
5  Problem Statement:
6
7  Prompt the user to enter a float number.
8  Use the round() function to round the number to 2 decimal places.
9  Print the original number and the rounded number.
10 Use the len() function to find the length of a string entered by the user and print the result.
11 """
12 #code Problem 1
13 inpt = float(input("Enter a float number: "))
14 inpt_rounded = round(inpt, 2)
15 print(f"Original number: {inpt}\nRounded number: {inpt_rounded}")
16 print(f"Length of the number entered by user :: {len(str(inpt))}")
17

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
34.4223223
● PS C:\Users\amjad\documents\Python-JP> python class-2.py
Enter a float number: 34.423332
Original number: 34.423332
Rounded number: 34.42
Length of the number entered by user :: 9
```

class-2.py X

C: > Users > amjad > Documents > Python-JP > class-2.py > ...

```
17
18 # problem 2
19 """
20 Problem Statement:
21
22 Prompt the user to enter a sentence.
23 Convert the entire sentence to uppercase.
24 Convert the entire sentence to lowercase.
25 Capitalize the first word of the sentence.
26
27 Print each of these modified sentences.
28 """
29
30 input_sentence = input("Enter a sentence: ")
31 print(f"Uppercase: {input_sentence.upper()}")
32 print(f"Lowercase: {input_sentence.lower()}")
33 print(f"Capitalized: {input_sentence.split(" ")[0].capitalize()}")
34
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
PS C:\Users\amjad\documents\Python-JP> python class-2.py
Enter a sentence: Hi My name is Amjad Ali Khan and im learning Python
Uppercase: HI MY NAME IS AMJAD ALI KHAN AND IM LEARNING PYTHON
Lowercase: hi my name is amjad ali khan and im learning python
Capitalized: Hi
```

```
class-2.py X
C: > Users > amjad > Documents > Python-JP > class-2.py > ...

35 # problem 3
36 """
37 Problem Statement:
38
39 Prompt the user to enter a sentence.
40 Ask user to replace the word
41 ask user to replace the word with
42
43 Print the modified sentence
44 """
45
46 input_sentence = input("Enter a sentence: ")
47 input_word = input("Enter the word to replace: ")
48 input_replace = input("Enter the word to replace with: ")
49 print(f"Original sentence: {input_sentence}")
50 print(f"Modified sentence: {input_sentence.replace(input_word, input_replace)}")
51

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS C:\Users\amjad\documents\Python-JP> python class-2.py
Enter a sentence: Hi my name is Amjad Ali Khan
Enter the word to replace: name
Enter the word to replace with: fullname
Original sentence: Hi my name is Amjad Ali Khan
Modified sentence: Hi my fullname is Amjad Ali Khan
```

```
53 # problem 4
54 """
55 Write a program that:
56 Asks the user to enter their age.
57 Adds 10 to their age.
58
59 Prints a message saying "In 10 years, you will be X years old." where X is the new age.
60 """
61
62 input_age = int(input("Enter your age: "))
63 new_age = input_age + 10
64 print(f"In 10 years, you will be {new_age} years old.")
65

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS C:\Users\amjad\documents\Python-JP> python class-2.py
Enter your age: 23
In 10 years, you will be 33 years old.
```

```

66 # problem 5
67 """
68 Write a program that:
69
70 Asks the user to enter their full name.
71 Converts the full name to uppercase and prints it.
72 Asks the user for their favorite number.
73 Multiplies the number by 2, converts it to a string, and concatenates it to a message.
74
75 Prints the message "Your favorite number multiplied by 2 is X.", where X is the new number.
76 """
77
78 input_name = input("Enter your full name: ")
79 input_number = int(input("Enter your favorite number: "))
80 print(f"Your name in uppercase: {input_name.upper()}")
81 new_number = input_number * 2
82 print(f"Your favorite number multiplied by 2 is {new_number}.")
83
84

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Users\amjad\documents\Python-JP> python class-2.py
Enter your full name: Amjad Ali khan
Enter your favorite number: 33
Your name in uppercase: AMJAD ALI KHAN
Your favorite number multiplied by 2 is 66.

```

```

84 # problem 6
85 """
86 Problem: Create a small program that asks the user for their first name and last name,
87 converts them to uppercase,
88 replaces spaces with hyphens
89 and calculates the length of their full name.
90
91 print 3 variables i.e
92 print("Your full name in uppercase is: " + full_name_upper)
93 print("Modified sentence: " + modified_sentence)
94 print("The length of your full name is: " + str(full_name_length))
95
96 NOTE: Concepts Covered: input(), string methods (upper, replace), len(), print()
97 """
98 input_first_name = input("Enter your first name: ")
99 input_last_name = input("Enter your last name: ")
100 full_name = input_first_name + " " + input_last_name
101 full_name_upper = full_name.upper()
102 modified_sentence = full_name.replace(" ", "-")
103 full_name_length = len(full_name)
104 print("Your full name in uppercase is: " + full_name_upper)
105 print("Modified sentence: " + modified_sentence)
106 print("The length of your full name is: " + str(full_name_length))
107
108

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Users\amjad\documents\Python-JP> python class-2.py
Enter your first name: Amjad
Enter your last name: Ali Khan
Your full name in uppercase is: AMJAD ALI KHAN
Modified sentence: Amjad-Ali-Khan
The length of your full name is: 14

```

```

108 # problem 7
109 """
110 Problem: Ask the user to input two numbers.
111 Calculate their average
112 and print the average rounded to 2 decimal places.
113
114 NOTE: Concepts Covered: round(), input(), print(), type casting
115 """
116
117 input_num1 = float(input("Enter the first number: "))
118 input_num2 = float(input("Enter the second number: "))
119 average = (input_num1 + input_num2) / 2
120 print(f"The average of the two numbers is {round(average, 2)}")
121

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

• PS C:\Users\amjad\documents\Python-JP> python class-2.py
Enter the first number: 45
Enter the second number: 43
The average of the two numbers is 44.0

```

```

122 # problem 8
123 """
124 Problem: Ask the user to input a sentence.
125 Replace all spaces with underscores
126 and split the sentence into words.
127
128 NOTE: Concepts Covered: replace(), split(), input(), print()
129 """
130
131 input_sentence = input("Enter a sentence: ")
132 modified_sentence = input_sentence.replace(" ", "_")
133 words = input_sentence.split(" ")
134 print(f"Modified sentence: {modified_sentence}")
135 print(f"Words: {words}")
136

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

• PS C:\Users\amjad\documents\Python-JP> python class-2.py
Enter a sentence: hi this is problem number 8
Modified sentence: hi_this_is_problem_number_8
Words: ['hi', 'this', 'is', 'problem', 'number', '8']

```